

Question	Answer	Marks
2(a)	a body continues at (rest or) constant velocity unless acted upon by a resultant force	B1
2(b)(i)1.	from 0–2 s, distance = $\frac{1}{2} \times 2 \times 6.8$ (= 6.8 m) and from 2–3 s, distance = $\frac{1}{2} \times 1 \times 3.4$ (= 1.7 m)	C1
	magnitude of displacement = 5.1 m	A1
	direction of displacement is down(wards)	B1
2(b)(i)2.	$(\Delta E) = mg\Delta h$ or $(E) = mgh$ or $(E) = Wh$	C1
	$(\Delta)E = 15 \times 5.1$ = (–) 77 J	A1
2(b)(ii)	$a = (v - u) / t$ or $a = \text{gradient}$ or $a = dv/dt$	C1
	$a = 3.4 \text{ ms}^{-2}$	A1
2(b)(iii)	$T - W = ma$ or $T - mg = ma$	C1
	$T = 15 + (15/9.81) \times 3.4 = 20 \text{ N or } 20.2 \text{ N}$	A1
2(b)(iv)	$E = F / A\epsilon$ or $E = \sigma / \epsilon$ <u>and</u> $\sigma = F / A$	C1
	$\epsilon = 20 / (2.8 \times 10^{-5} \times 1.7 \times 10^{11})$	C1
	= 4.2×10^{-6}	A1
2(b)(v)	block is in equilibrium/has no resultant force	B1
	block could be stationary (or have constant velocity/speed) (so no, not possible to deduce)	B1

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2(c)	mass is the property (of a body/object) resisting changes in motion	D1